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Title: Flywheel-Crank Mechanism

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Introduction

- The flywheel–crank mechanism converts rotary motion into linear reciprocating motion and vice-versa.
- It consists of a flywheel, crankshaft, connecting rod, and slider working together.
- The flywheel stores energy and reduces speed fluctuations, ensuring smooth operation.
- This mechanism is widely used in IC engines, pumps, compressors, and other mechanical systems.

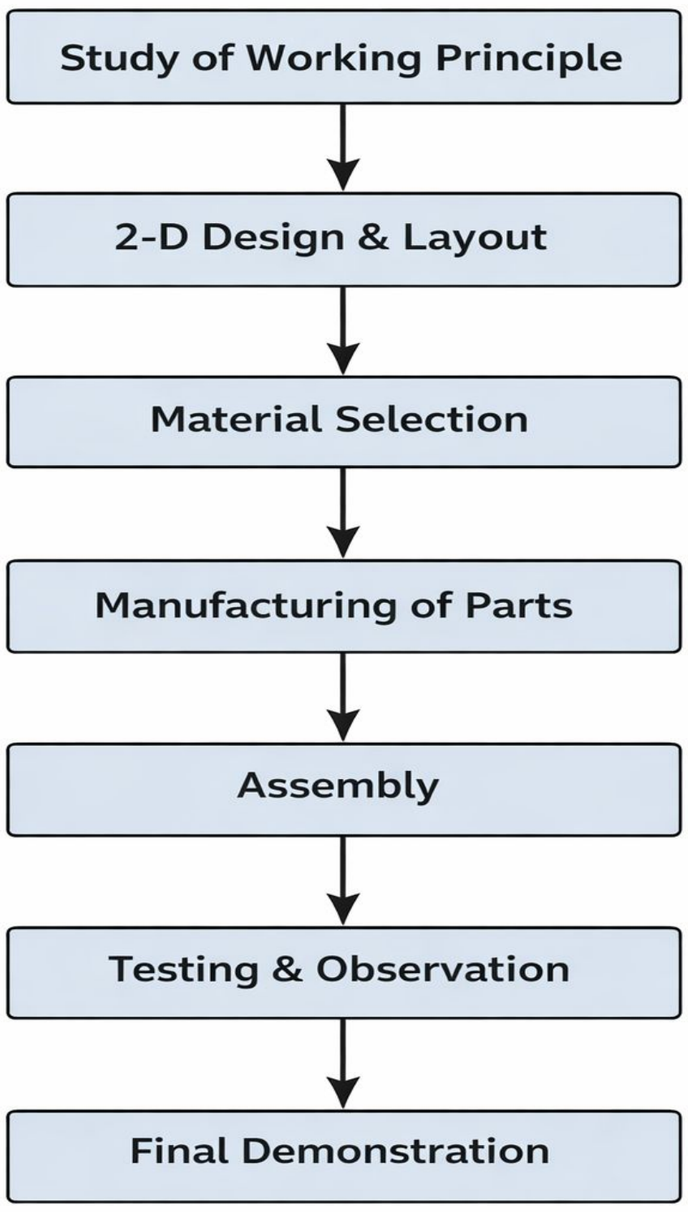
Problem Definition

- Many mechanical applications require efficient conversion between rotary and reciprocating motion.
- Direct motion conversion without energy storage leads to uneven motion, vibration, and speed fluctuations.
- A reliable mechanism is required to ensure smooth motion transmission and continuous operation.
- The flywheel–crank mechanism addresses this problem by providing effective motion conversion with energy storage.

Objectives

- To study the working principle of the flywheel–crank mechanism.
- To understand the conversion of rotary motion into reciprocating motion.
- To analyze the role of the flywheel in reducing speed fluctuations.
- To fabricate and demonstrate a working model of the mechanism.

Methodology



Flowchart showing step-tep-step methodology followed for fabrication and testing of the flywheel–crank mechanism.

Tools used

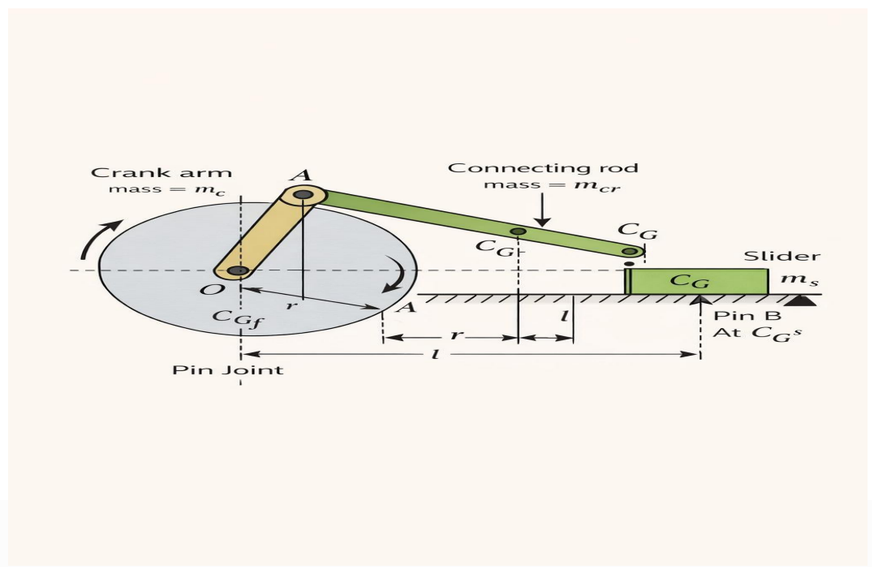
- Lathe Machine
- Drilling Machine
- Milling Machine
- Bench vice
- Measuring instruments (vernier caliper, steel scale)
- Spanners and fasteners

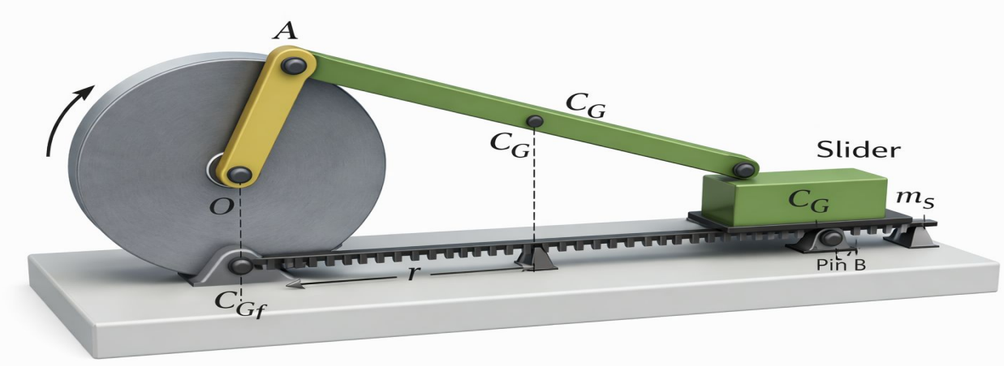






2D and 3D Drawings





Conclusions

- The flywheel–crank mechanism successfully converts rotary motion into reciprocating motion.
- The flywheel stores rotational energy and minimizes speed fluctuations during operation.
- Proper design and alignment of the crankshaft and connecting rod improved smoothness and efficiency.
- The mechanism closely resembles its application in real systems such as IC engines and reciprocating pumps.
- This project enhanced understanding of kinematics, manufacturing processes, and mechanical assembly techniques.

Visuals

